

Sailab Banik

Chennai, Tamil Nadu, India | +91 87XXX XXXXX | sailabbanik24@gmail.com | sailab-banik-portfolio.vercel.app
linkedin.com/in/sailabbanik-24 | github.com/sailab-banik | medium.com/@sailabbanik24

SUMMARY

Software Engineer with 3+ years building production AI systems and the full-stack products around them, specializing in **Python, FastAPI, React, and LLM systems**. Experienced in **RAG pipelines, semantic caching, and document intelligence**, with end-to-end ownership across architecture, CI/CD, and production delivery.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, HTML, CSS

AI/ML: Generative AI, RAG Pipelines, Large Language Models (LLMs), Vision-Language Models (VLMs), Semantic Caching, Context Engineering, LangChain, Ollama, OpenAI API, Redis Iris (LangCache, Agent Memory), Vector Databases (Chroma)

Backend: FastAPI, REST APIs, PostgreSQL, Redis, SQLAlchemy, Pydantic, OAuth2 / SSO (Microsoft Entra ID, MSAL), JWT Authentication

Frontend: React.js, Next.js, Redux Toolkit, Material UI, Vite, React Testing Library, Vitest, MSW

DevOps & Tools: Docker, Docker Compose, CI/CD Pipelines, Azure DevOps, Git, GitLab

WORK EXPERIENCE

Software Engineer II

ZF Group

Oct 2024 – Present

Chennai, India

- Built the AI layer of **PrimeAI**, an internal document processing platform adopted by **5+ business teams**: annotation-driven extraction, user-defined model creation, and the retrieval that assembles context before each model call — cutting manual review effort by **60%** and saving **2,000+ operational hours per quarter**.
- Routed extraction between **locally served models on Ollama**, for documents that cannot leave internal infrastructure, and frontier APIs where accuracy required them, closing most of the gap with retrieval rather than a larger model.
- Led frontend development and owned end-to-end product flows across UI, API, and database layers; designed **role- and group-based access control** from the ground up, enabling rollout to departments with strict data-access requirements.
- Drove CI/CD improvements across the team, reducing release cycle time by **40%** and production hotfix incidents by **35%**; introduced frontend test suites using **React Testing Library and Vitest**, reducing manual testing effort by **60%**.

Graduate Engineer Trainee

ZF Group

Aug 2023 – Sep 2024

Chennai, India

- Stabilized a business-critical legacy platform by resolving **10+ high-priority production bugs**, reducing user-reported issues by **45%**.
- Led the **security hardening pass** across internal applications ahead of audit, remediating injection and input-validation findings and carrying every application through review.
- Optimized operations scheduling workflows, improving throughput by **30%** and earning the **Budding Star Award** for measurable business contribution within the first year.

PROJECTS

ContextDesk Lab: Context Engineering Playground for Redis Iris

Python, FastAPI, Redis Iris, Next.js, TypeScript, Docker

- Built an AI support console where every request returns a response *and* an execution record: its path through cache, memory, and retrieval, the tokens each source contributed, and what was dropped and why — orchestrating **Redis Iris LangCache, Agent Memory, and Context Retriever**.
- Cut repeat-question latency from **4.65s to 917ms with 0 tokens billed** by answering semantically equivalent prompts from **Lang-Cache**, scoring every lookup below the decision threshold so a miss still reports its similarity; every component ships a stub, under **50 offline tests**.

QueueLens: Distributed Job Queue with Live Dashboard

Python, FastAPI, Redis, PostgreSQL, Next.js, Docker

- Built a producer-consumer job queue: a **FastAPI** producer persists jobs in **PostgreSQL** and hands job IDs to background workers over a **Redis** list (LPUSH/BRPOP), with a **Next.js** dashboard tracing every job through its lifecycle live.
- Implemented **retries with exponential backoff**, Redis-backed **per-client rate limiting** that holds across API processes, and a queue cap returning 429/503 with Retry-After; ordered state commits before enqueueing to prevent API-worker race conditions.

Advanced RAG Pipeline for 100+ Page Documents

Python, FastAPI, LangChain, Ollama, Qwen3, Chroma

- Built a pipeline converting unstructured **100+ page PDFs** into structured JSON from user-defined schemas, running fully on local infrastructure with **Ollama-hosted Qwen3** and keeping sensitive data within the environment.
- Designed a **dual-path RAG architecture** with separate text and table vector stores, structure-aware chunking, and cross-encoder reranking, improving retrieval precision by **~40%** over a single-store baseline.
- Implemented runtime validation using **dynamically generated Pydantic models** and an **LLM-driven repair loop**, improving valid-JSON responses from **~75% to 100%**; exposed it through a modular REST API with idempotent ingestion.

EDUCATION

ICFAI University Tripura

B.Tech in Computer Science and Engineering – **CGPA: 9.64**

Aug 2019 – Jul 2023

Agartala, India

CERTIFICATIONS & AWARDS

Budding Star Award — ZF Group, Recognition for Business Impact

2024

System Design Cohort — ChaiCode: distributed systems, load balancing, caching, sharding, queues

Credential | Jun 2026

AI Engineer Core Track (LLM Engineer) — Udemy: production LLM apps, RAG, agentic workflows, fine-tuning

Credential | May 2026